

BIO 102: Practice Examination

Answer all questions. Illustrate your answers with clear reasoning/labelled diagrams where appropriate.

1. A rare genetic disorder, "Syndrome X," is caused by a recessive allele (x) at an autosomal locus. In a population of 10,000 people that is in Hardy-Weinberg Equilibrium for this locus, the frequency of the dominant wild-type allele (X) is 0.99.

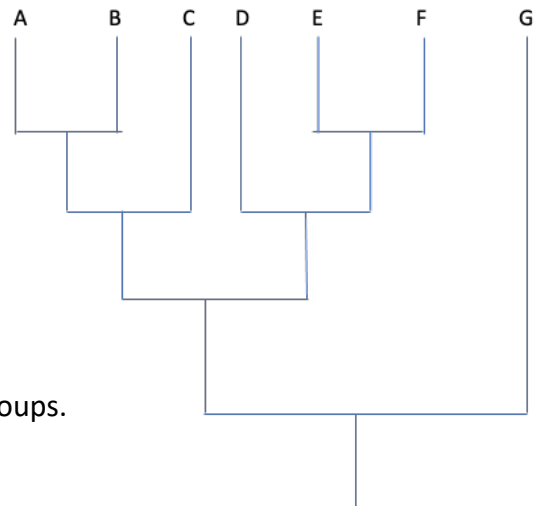
- Calculate the frequency of the recessive allele (x). (1 mark)
- How many individuals in this population would you expect to be carriers for Syndrome X (i.e., heterozygous)? (3 marks)
- If a child is born with Syndrome X, what must be the genotypes of both parents? Does this observation violate the assumptions of the Hardy-Weinberg Equilibrium for this specific family? Explain. (2 marks)

2. In maize, two genes located on the same chromosome are studied: the gene for seed colour (Purple, C, is dominant over white, c) and the gene for seed shape (Smooth, S, is dominant over wrinkled, s). A dihybrid plant is crossed with a homozygous recessive plant (test cross). The offspring from this test cross are as follows:

Phenotype	Number of Offspring
Purple, Smooth	380
Purple, Wrinkled	115
White, Smooth	105
White, Wrinkled	400
Total	1000

- Based on the offspring numbers, are the genes for seed colour and seed shape linked or assorting independently? Provide a reason for your answer. (1 mark)
 - Identify the parental and recombinant phenotypes. (2 marks)
 - Calculate the recombination frequency between the C and S genes. Show your working. (2 marks)
 - What is the map distance between these two genes in centiMorgans (cM)? (1 mark)
3. The diagram below shows a phylogenetic tree for seven extant species (A-G).
- Identify three sister groups and explain your reasoning. (1 mark)

b. A researcher discovers that Species G has a simple, single-chambered stomach, while all other species (A-F) have complex, multi-chambered stomachs. Assuming the simple stomach is the ancestral trait, indicate the most likely point on the tree where the complex stomach evolved. (2 marks)



c. What are nodes and branches on a phylogenetic tree? (2 marks)

d. Indicate the most recent common ancestor of all the groups. (1 mark)

4. Describe with examples where appropriate

a. Evolution (2 marks)

b. Exaptation (2 marks)

c. Stabilizing Selection (2 marks)

5. Before the Industrial Revolution in England, the peppered moth (*Biston betularia*) was predominantly light-coloured with dark spots, which camouflaged it against lichen-covered trees. Dark-coloured (melanic) moths were very rare. During the Industrial Revolution, soot from factories darkened the tree trunks and killed the lichens. By the late 19th century, the dark-coloured form had become the most common in polluted areas. With the introduction of Clean Air Acts in the mid-20th century, the light-coloured form has become common again in those same areas.

Using this example, comment on the following statements:

a. Evolution is a change in allele frequencies over time. How does this example demonstrate this definition? (2 marks)

b. Evolution is not "goal-directed". How does the reversal of the moth populations after the Clean Air Acts illustrate this? (2 marks)

c. Fitness is an interaction between an organism and its environment. Explain how the fitness of the dark moth changed over time in the polluted region. (2 marks)